UL TEST REPORT AND PROCEDURE

Standard:	UL 62368-1, 4th Ed, 2025-07-31 (Audio/Video, information and communication technology equipment Part 1: Safety requirements) CAN/CSA C22.2 No. 62368-1:25, 4th Ed, 2025-07-31 (Audio/video, information and communication technology equipment Part 1: Safety requirements)
Certification Type:	Listing
CCN:	AZOT, AZOT7 (Audio/video, Information and Communication Technology Equipment)
Complementary CCN:	N/A
Product:	Embedded Box PC
Model:	ACS-330
Rating:	1) 9-36Vdc, 9-2.2A 2) 9-36Vdc, 5.2-1.3A or 12Vdc, 4A (See general product information and other remarks for details)
Applicant Name and Address:	APLEX TECHNOLOGY INC 15TH FL-1 NO 186 JIAN YI RD ZHONGHE DISTRICT NEW TAIPEI 235 TAIWAN

This is to certify that representative samples of the products covered by this Test Report have been investigated in accordance with the above referenced Standards. The products have been found to comply with the requirements covering the category and the products are judged to be eligible for Follow-Up Service under the indicated Test Procedure. The manufacturer is authorized to use the UL Mark on such products which comply with this Test Report and any other applicable requirements of UL LLC ('UL') in accordance with the Follow-Up Service Agreement. Only those products which properly bear the UL Mark are considered as being covered by UL's Follow-Up Service under the indicated Test Procedure.

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Supporting Documentation

The following documents located at the beginning of this Procedure supplement the requirements of this Test Report:

A. Authorization - The Authorization page may include additional Factory Identification Code markings.

B. Generic Inspection Instructions -

i. Part AC details important information which may be applicable to products covered by this Procedure. Products described in this Test Report must comply with any applicable items listed unless otherwise stated in the body of this Test Report.

ii. Part AE details any requirements which may be applicable to all products covered by this Procedure. Products described in this Test Report must comply with any applicable items listed unless otherwise stated in the body of each Test Report.

iii. Part AF details the requirements for the UL Certification Mark which is not controlled by the technical standard used to investigate these products. Products are permitted to bear only the Certification Mark(s) corresponding to the countries for which it is certified, as indicated in each Test Report.

Product Description

The equipment, model ACS-330, is audio/video, information and communication technology equipment.

Electrical components are mounted on PCB, which is enclosed by metal enclosure.

The product is intended to be installed on a desktop or other flat surface horizontally or mounted on the wall or Din rail.

The I/O ports of mainboard are as follow: DC input terminal block connecter x 1, USB 2.0 port x 4, LAN ports x 2, HDMI port x 1, DP port x 1, GPIO port x 1, COM port x 2.

The I/O ports o Expansion card are as follow:

- TB-648: USB 2.0 port x 4, USB 3.0 port x 4.
- TB-636: USB 2.0 port x 2, LAN ports x 2.
- TB-635: COM port x 4 and GPIO x 1.

There are four construction, see enclosure miscellaneous ID 7-01 for details.

Model Differences

N/A

Test Item Particulars

Product group ..... :	end product
Classification of use by ..... :	Ordinary person Children likely present
Supply connection ..... :	not mains connected: ES1
Supply tolerance ..... :	None
Supply connection - type ..... :	not directly connect to the mains
Considered current rating of protective device ... :	N/A

Equipment mobility ..... :	movable wall/ceiling-mounted
Overvoltage category (OVC) ..... :	not directly connect to the mains
Class of equipment ..... :	Class III
Special installation location ..... :	N/A
Pollution degree (PD) ..... :	PD 2
Manufacturer's specified Tma ..... :	60 °C
IP protection class ..... :	IPX0
Power systems ..... :	not AC mains
Altitude during operation (m) ..... :	5000 m
Altitude of test laboratory (m) ..... :	2000 m or less
Mass of equipment (kg) ..... :	Max. 1.63 kg

Technical Considerations

- The Risk Group of a lamp or lamp system (including LEDs) is : Exempt
- The following are available from the Applicant upon request : Installation (Safety) Instructions / Manual
- The following circuit locations (with circuit/schematic designation) were investigated as a PS2: all accessible output ports.
- The equipment is intended to be supplied by approved external power source (UL listed / IEC 62368-1) which output is complied with ES1, output rating 9-36Vdc, 9-2.2A, ambient temperature 60°C minimum and altitude of operation up to 5000m. (for construction A).
- The equipment is intended to be supplied by approved external power source (UL listed / IEC 62368-1) which output is complied with ES1, output rating 9-36Vdc, 5.2-1.3A or 12Vdc, 4A, ambient temperature 60°C minimum and altitude of operation up to 5000m. (for construction B, C, D).

Additional Information

All tests were conducted under normal load conditions as below, if not specified elsewhere.

Maximum normal load:
The EUT connected to a monitor via HDMI/DP port, run programs, RJ-45 port connected to a simulator with highest speed data transmission, each front USB2.0 port loaded to 0.5A (total 4 provided), and operated continuously with following expansion card condition:
- for construction A: with expansion card TB-648, each rear USB3.0 port loaded to 0.9A (total 4 provided), each rear USB2.0 port loaded to 0.5A (total 4 provided).
- for construction B: with expansion card TB-636, each rear USB2.0 port loaded to 0.5A (total 2 provided).
- for construction C: with expansion card TB-635.
- for construction D: without expansion card.

Unless other specified, all test were performed on construction A to represent other similar construction.

Markings and Instructions

Clause Title	Marking or Instruction Details
Inter-connecting cables - External detachable	Responsible company’s name and Part number (Marking or Instruction)
Equipment identification marking – Manufacturer identification	Listee’s or Recognized Company’s name, Trade Name, Trademark or File Number

Equipment identification marking – model identification	Model Number
Equipment rating marking – ratings	Input Ratings (voltage, frequency/dc, current/power) (Optional)
Battery replacement	Provided (on equipment, in instruction manual) CAUTION - Risk of fire or explosion if the battery is replaced by an incorrect type. Dispose of Used Batteries According to the Instructions
Instruction/Installation/Safety	For operating / instruction / safety manual / user's manual might be provided by following ways: (1) hardcopy; or (2) via CD-ROM or hard disk drive or internet or equivalent type of soft media. If the instruction / installation / safety manual hardcopy is not shipped with unit, a declaration shall be provided to users and they can access electronic instructions from CD-ROM or hard disk drive or internet or equivalent type of soft media. - If the power adapter is not shipped with unit, instruction / installation / safety manual shall have descriptions as below in quotation marks or equivalent, and the power adapter shall be UL Listed unit: (1) "This unit is intended to be supplied by a power adapter suitable for use at 60 degree C min., altitude of operation up to 5,000 m (16,404 feet) whose output is rated 1) 9-36Vdc, 9-2.2A (For equipment with expansion card TB-648) or 2) 9-36Vdc, 5.2-1.3A or 12Vdc, 4A (For equipment with expansion card TB-635 or TB-636, or without expansion card)" or (2) “Use only the Aplex provided AC adapter approved for use with this device”.
Protective earthing is used as a safeguard	"Ensure to connect the power cord to a socket-outlet with earthing connection" or See Enclosure Manual for equivalent details.
Wall Mount / Panel Mount instruction	See Enclosure / Manual for details.
Special Instructions to UL Representative N/A	

BD1.0	TABLE: Production-Line Testing Requirements					
BD1.1	Electric Strength Test Special Constructions – Refer to Generic Inspection Instructions, Part AC for further information.					
Model	Component	Removable parts	Test probe location	Test V rms	Test V dc	Test Time, s
N/A	--	--	--	--	--	--
BD1.2	Earthing Continuity Test Exemptions – This test is not required for the following models:					
	All models.					
BD1.3	Electric Strength Test Exemptions – This test is not required for the following models:					
	All models.					
BD1.4	Electric Strength Test Component Exemptions – The following solid-state components may be disconnected from the remainder of the circuitry during the performance of this test.					
	N/A					

BE1.0	Sample and Test Specifics for Follow-Up Tests at UL				
Model	Component	Material	Test	Sample (s)	Test Specifics
N/A	--	--	--	--	--

4.1.2	TABLE: List of critical components					Pass
Object / part No.	Manufacturer/ trademark	Type / model	Technical data	Product Category CCN(s)	Mark(s) of conformity	Supplement ID
01. Label	Interchangeable	Interchangeable	Min. 100 degree C, if maximum surface temperature not specified.	PGDQ2 or PGJI2	UL	
01a. Label (Alternate)	--	--	Required Markings molded in Enclosure or Permanently Silk-screened on surface of Enclosure.	--	--	
02. Metal enclosure	--	--	Aluminum alloy, min 1.0mm thickness. See Diagram 04-01 for overall dimension details.	--	--	
03. Wall mount kit (Optional)	--	--	SECC, min 1.0mm thickness. See Diagram 04-02 for overall dimension details.	--	--	
04. DIN-rail kit (Optional)	--	--	Steel, min 1.2mm thickness. See Diagram 04-03 for overall dimension details.	--	--	
05. Terminal block (Input- Receptacle) (J4) (on main board)	Interchangeable	Interchangeable	Minimum 36V, minimum 9A, minimum 105 °C. Metal/Plastic Copper alloy pins housed in bodies of plastic rated V-1 minimum.	XCFR2	UL	
06. Terminal block (Input- Connector)	Interchangeable	Interchangeable	Minimum 36V, minimum 9A, minimum 105 °C.Metal/Plastic Copper alloy pins housed in bodies of	XCFR2	UL	

			plastic rated V-1 minimum. FW2. 28-12AWG, Cu type wire, 4.5 lb-in Torque.			
07. RTC battery (Lithium type) (connected to VBAT1 of main board)	Interchangeable	CR2477, CR-2477, CR2477*, CR-2477*	Maximum abnormal charging current 5mA min., reverse current protection by diode (D1) and resistor (R37) rated 1k ohm min.	BBCV2	UL	
08. IC current limiter (PU1, PU2) (on main board) (for USB2.0 port (USB1, USB2))	UPI SEMICONDUCTOR CORP	uP7534CMA5-20	Input voltage: 2.7-5.5 Vdc,Input current: 3.3-5.0A max., Current limit: 5.0 A max.	QVGS2	UL	
09. Capacitor (PCE7, C65, C346) (on main board)	Interchangeable	Interchangeable	Min. 105 °C	--	--	
10. IC current limiter (F1, F2, F3, F4) (on expansion card TB-648) (for USB ports)	GLOBAL MIXED-MODE TECHNOLOGY INC	G509H20DT11G	Input voltage: 2.7-5.5 Vdc, Output Continuous Rating: 2.1 A, Output Current Limit: 2.2-2.8 A	QVGS2	UL	
11. Thermal pad (for CPU of main board)	Interchangeable	Interchangeable	Min. V-2 or flammability level is ignored when small parts (4g max. or 1750mm^3 max.). See Diagram 04-04 for thermal pad location and dimension details.	QMFZ2	UL	
12. Printed Wiring Boards	Interchangeable	Interchangeable	Minimum V-1, minimum 130 °C	ZPMV2	UL	
13. Heatsink (for CPU)	--	--	Aluminum. See Diagram 04-05 for dimension details.	--	--	
14. Connectors and Receptacles (for	Interchangeable	Interchangeable	Metal/Plastic Copper alloy pins housed in	QMFZ2	UL	

internal ports) (Secondary ES1 Circuits)			bodies of plastic rated V-2 min.			
14a. Connectors and Receptacles (for internal ports) (Secondary ES1 Circuits) (Alternate)	Interchangeable	Interchangeable	--	ECBT2 or DUXR2	UL	
14b. Connectors and Receptacles (for internal ports) (Secondary ES1 Circuits) (Alternate)	--	--	Flammability level is ignored, when small parts (4g max. or 1750mm^3 max.)	--	--	
15. Connectors and Receptacles (for external ports) (Secondary ES1 Circuits)	Interchangeable	Interchangeable	Metal/Plastic Copper alloy pins housed in bodies of plastic rated V-1 min.	QMFZ2	UL	
15a. Connectors and Receptacles (for external ports) (Secondary ES1 Circuits) (Alternate)	Interchangeable	Interchangeable	--	ECBT2 or DUXR2	UL	
16. Interconnecting Cable (Optional)	Interchangeable	Interchangeable	Minimum 80 degree C, minimum 30V, maximum 3.05m long, VW-1 or FT-1 or better.	AVLV2 or ZPFW2	UL	
16a. Interconnecting Cable (Optional) (Alternate)	Interchangeable	Interchangeable	Maximum 3.05m long, type CMP, CMR, CMG, CM, CMX, CMUC, or CMH.	DUZX	UL	
16b. Interconnecting Cable (Optional) (Alternate)	Interchangeable	Interchangeable	With RJ-45 type connector	DUXR	UL	
17. Internal wiring (Secondary ES1 Circuits)	Interchangeable	Interchangeable	FEP, PTFE, PVC, TFE, neoprene, polyimide or	AVLV2	UL	

			VW-1 or FT-1, Min. 105 °C. Min. 60 V.			
18. Internal Plastic Part Materials (Optional)	Interchangeable	Interchangeable	Min. V-2, VTM-2 or HF-2.	QMFZ2	UL	
18a. Internal Plastic Part Materials (Optional) (Alternate)	--	--	Flammability level is ignored, when small parts (4g max. or 1750mm^3 max.) are mounted on V-1 class material.	--	--	
18b. Internal Plastic Part Materials (Optional) (Alternate)	--	--	Flammability level is ignored, when small parts (4g max. or 1750mm^3 max.) are separated from electrical parts (other than insulated wires and cables) by at least 13 mm of air or by a solid barrier of V-0 class material.	--	--	
19. External power adapter (for construction A)(Optional)	Interchangeable	Interchangeable	O/P: 9-36 Vdc, 9-2.2 A min., Tma: 60 °C min., Altitude: 5000m min.	QQJQ	UL	
20. External power adapter (for construction B to C) (Optional)	Interchangeable	Interchangeable	O/P: 9-36 Vdc, 5.2-1.3 A min., Tma: 60 °C min., Altitude: 5000m min.	QQJQ	UL	
20a. External power adapter (for construction B to C) (Optional) (Alternate)	Interchangeable	Interchangeable	O/P: 12 Vdc, 4 A min., Tma: 60 °C min., Altitude: 5000m min.	QQJQ	UL	
21. Power supply cord (Optional)	Interchangeable	Interchangeable	Detachable. Max. 2.4m (14.76ft) and min. 1.5m long; Type NISP-2,	ELBZ	UL	

			NISPE-2, NISPT-2, SP-2, SPE-2, SPT-2, SP-3, SPE-3, SV, SVE, SVO, SVOO, SVT, SVTO or SVTOO, No. 18 AWG minimum. One end terminates in NEMA 5-15P or NEMA 6-15P for Class I or one end terminates in NEMA 1-15P or NEMA 2-15P for Class II, the other end with an appliance coupler.			
21a. Power supply cord (Optional) (Alternate)	Interchangeable	Interchangeable	Detachable, maximum 4.5 m long, minimum 1.5 m long. Type S, SE, SJ, SJE, SJO, SJOO, SJT, SJTO, SJTOO, SO, SOO, ST, STO, STOO PPE or W, no. 18AWG minimum. One end terminates in NEMA 5-15P or NEMA 6-15P for Class I or one end terminates in NEMA 1-15P or NEMA 2-15P for Class II, the other end with an appliance coupler.	ELBZ	UL	

Enclosures

Type	Supplement Id	Description
Photographs	03-01	External view -1_for all contstruction
Photographs	03-02	External view -2_for contstruction A
Photographs	03-03	External view -3_for all contstruction
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Photographs	03-06	Internal view -1_for contstruction A
Photographs	03-07	Internal view -2_for contstruction A
Photographs	03-08	Internal view -3_for all contstruction
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Manuals	06-01	User manual
Miscellaneous	07-01	Construction table